FIRMAR CORRESO ELECTRÓNICOS CON S/MIME USANDO THUNDERBIRD Y CERTIFICADOS PROPIOS

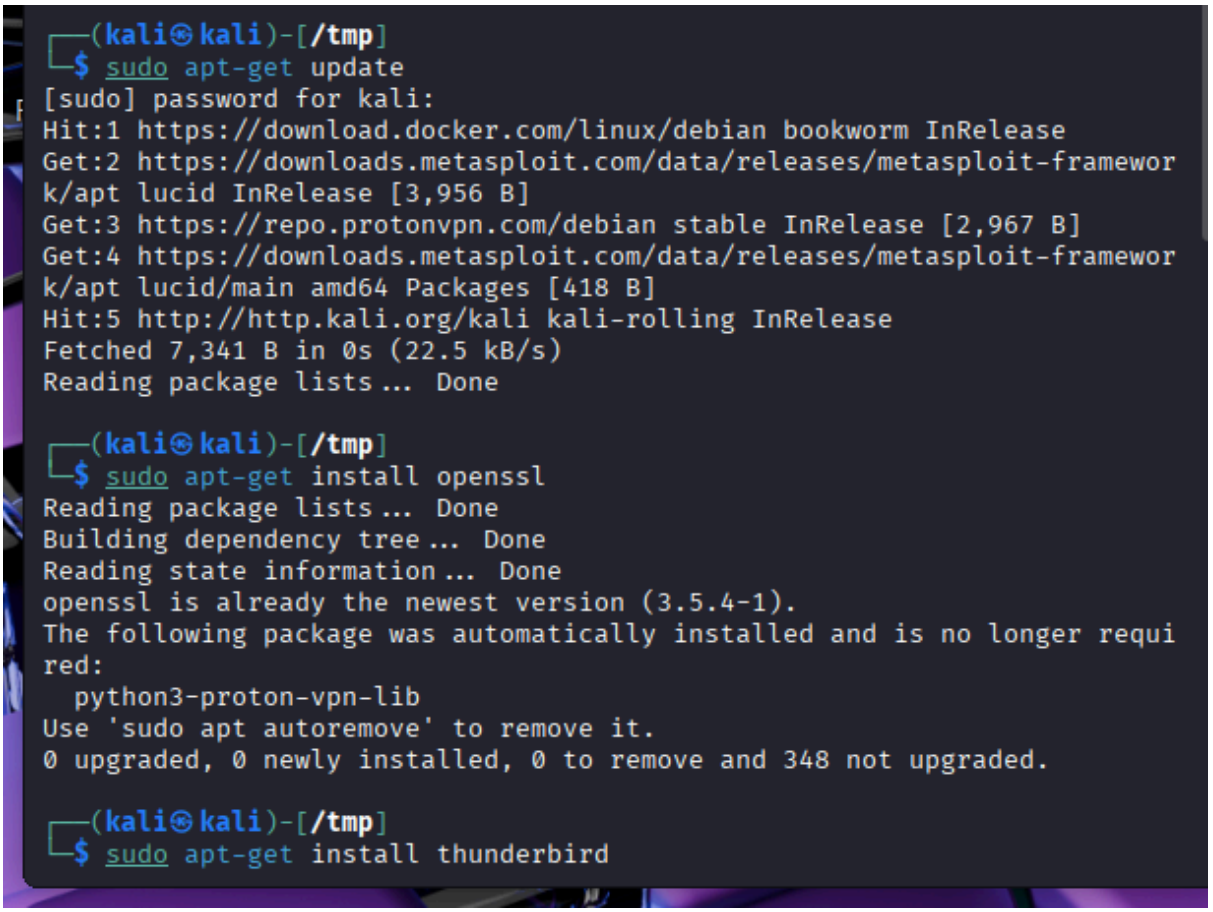
Instalación

Primero, necesitamos instalar las herramientas que utilizaremos para crear la CA y los certificados, para ello usaremos una máquina virtual de Ubuntu:

```
sudo apt-get update

sudo apt-get install openssl

sudo apt-get install thunderbird
```



```
┌──(kali㉿kali)-[/tmp]
└─$ sudo apt-get install thunderbird
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
thunderbird is already the newest version (1:140.4.0esr-1).
The following package was automatically installed and is no longer requi
red:
  python3-proton-vpn-lib
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 348 not upgraded.
```

2. Crear la autoridad de certificación (CA)

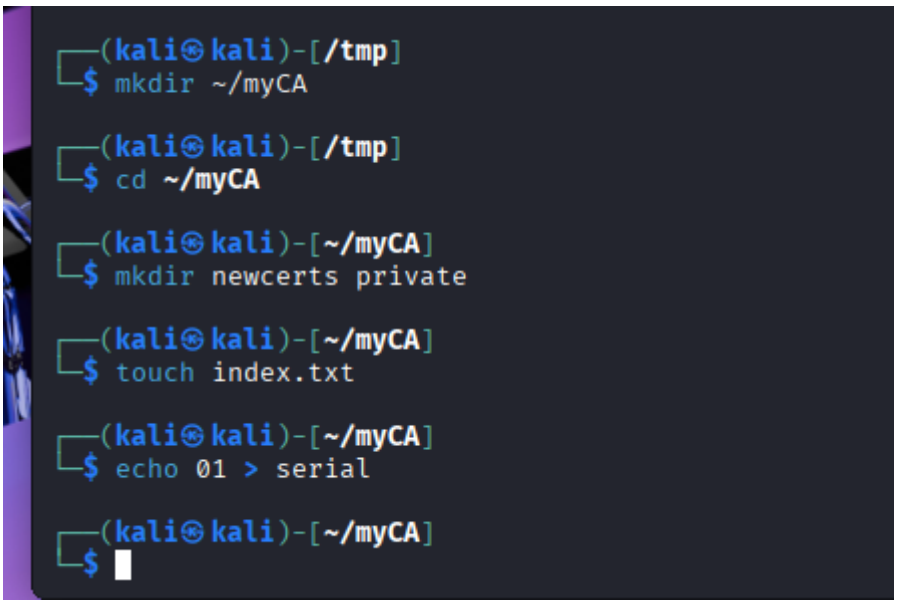
```
mkdir ~/myCA

cd ~/myCA

mkdir newcerts private

touch index.txt

echo 01 > serial
```



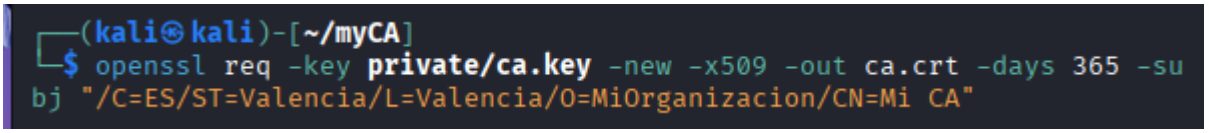
3. Generar la clave privada de la CA

```
openssl genpkey -algorithm RSA -out private/ca.key -pkeyopt rsa_keygen_bits:2048
```



4. Crear el certificado de la CA

```
openssl req -key private/ca.key -new -x509 -out ca.crt -days 365 -subj
"/C=ES/ST=Valencia/L=Valencia/O=MiOrganizacion/CN=Mi CA"
```



5. Crear y firmar certificados para S/MIME

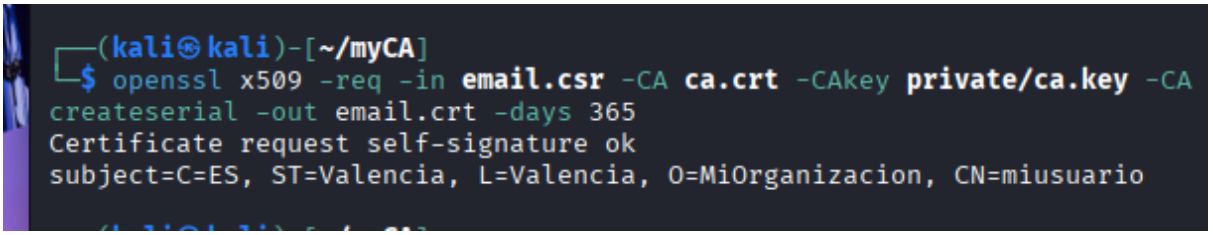
```
openssl genpkey -algorithm RSA -out private/email.key -pkeyopt rsa_keygen_bits:2048
```



6. Crear una CSR (solicitud de firma de certificado)

```
openssl req -new -key private/email.key -out email.csr -subj "/C=ES/ST=Valencia/L=Valencia/O=MiOrganizacion/CN=miusuario"

openssl x509 -req -in email.csr -CA ca.crt -CAkey private/ca.key -CAcreateserial -out email.crt -days 365
```

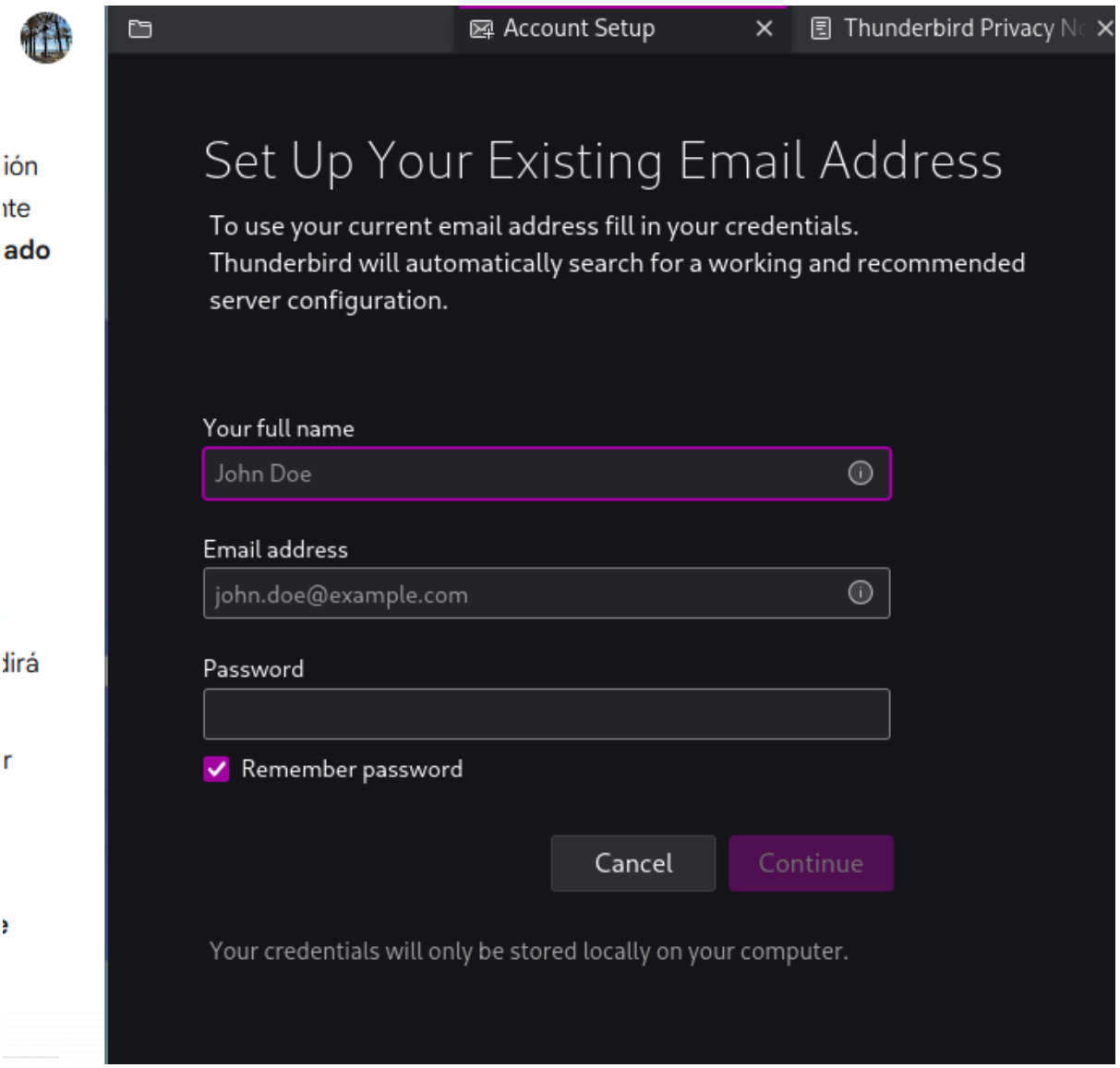


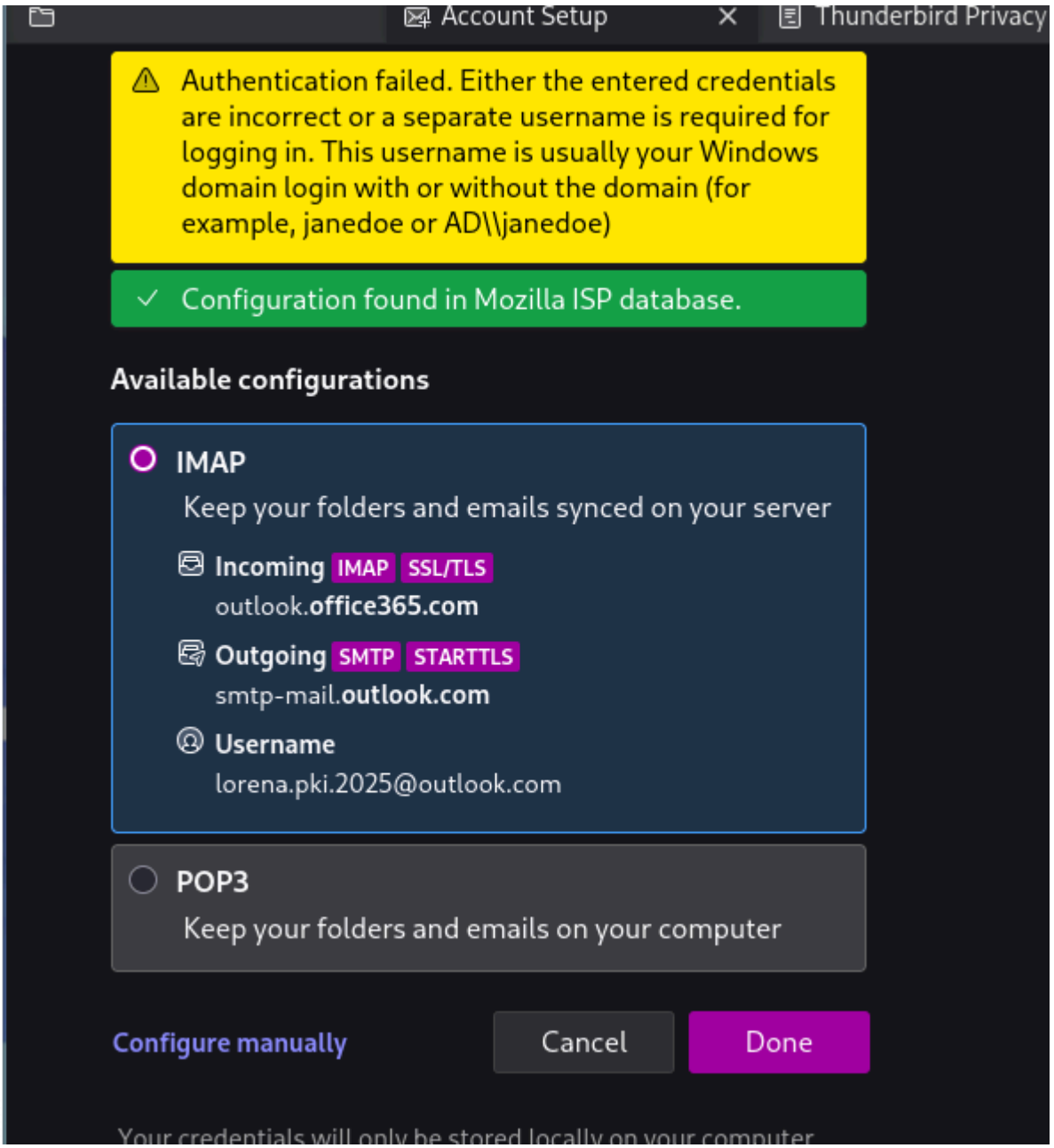
7. Convertir a formato PKCS#12

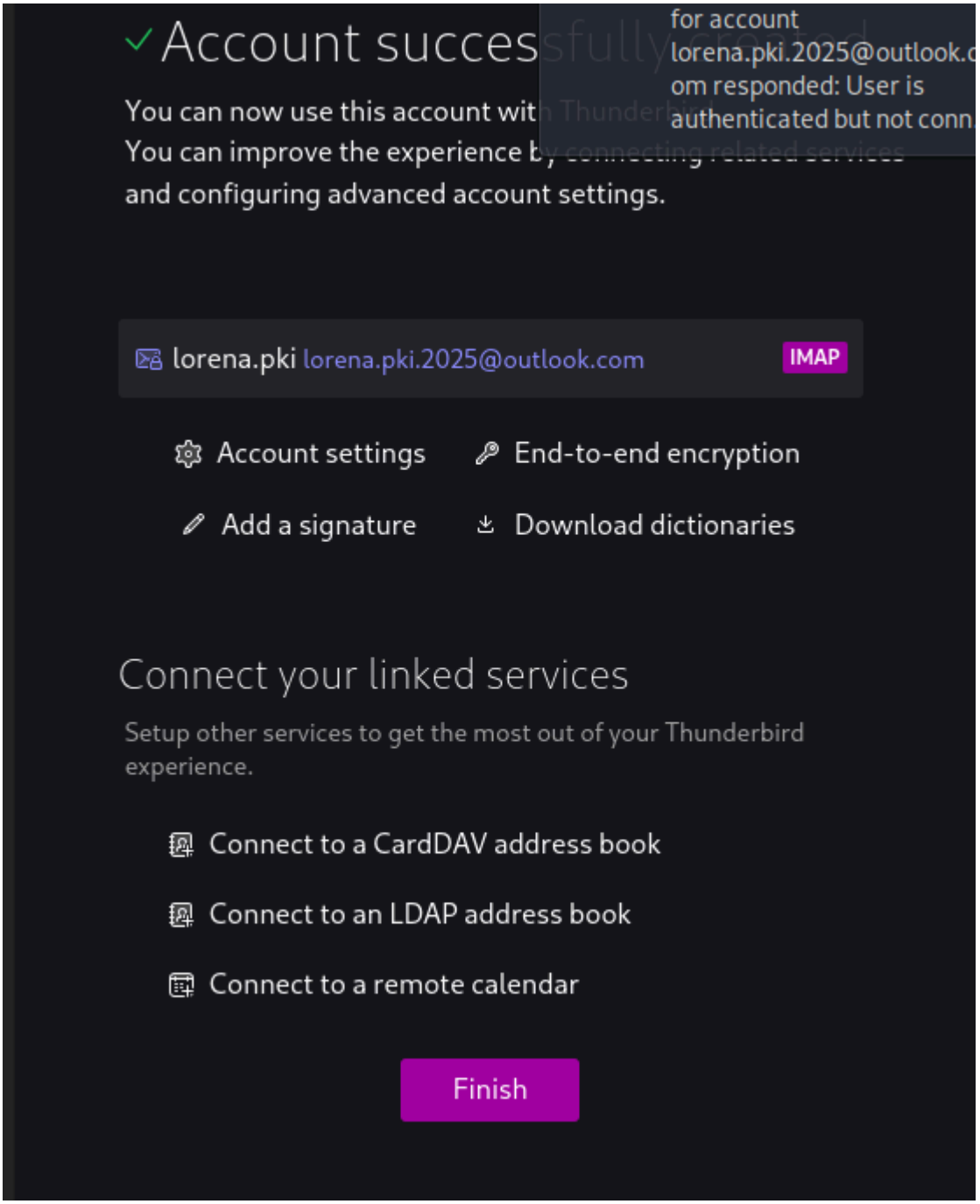
```
openssl pkcs12 -export -out email.p12 -inkey private/email.key -in email.crt -certfile ca.crt
```

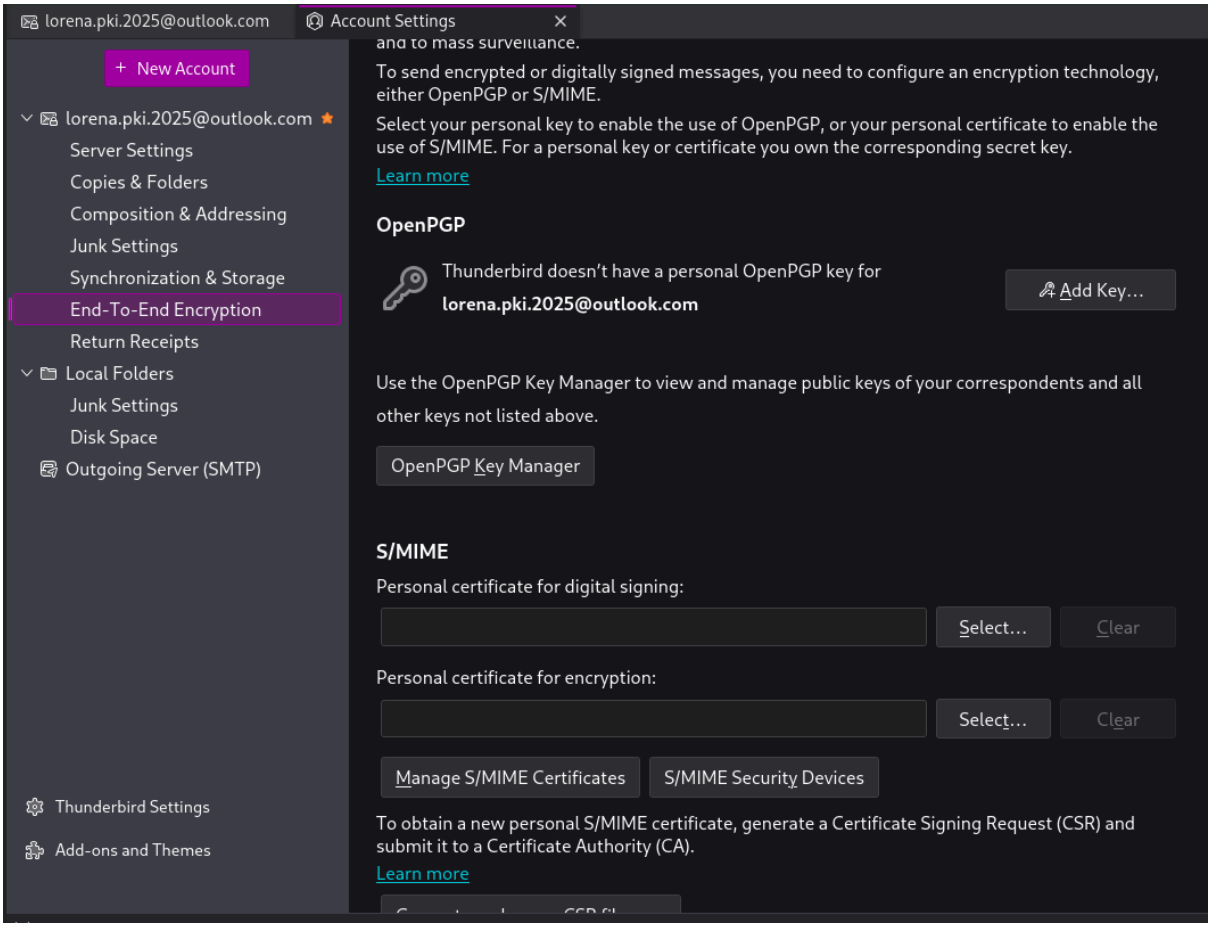
8. Configurar Thunderbird para usar S/MIME e importar el certificado

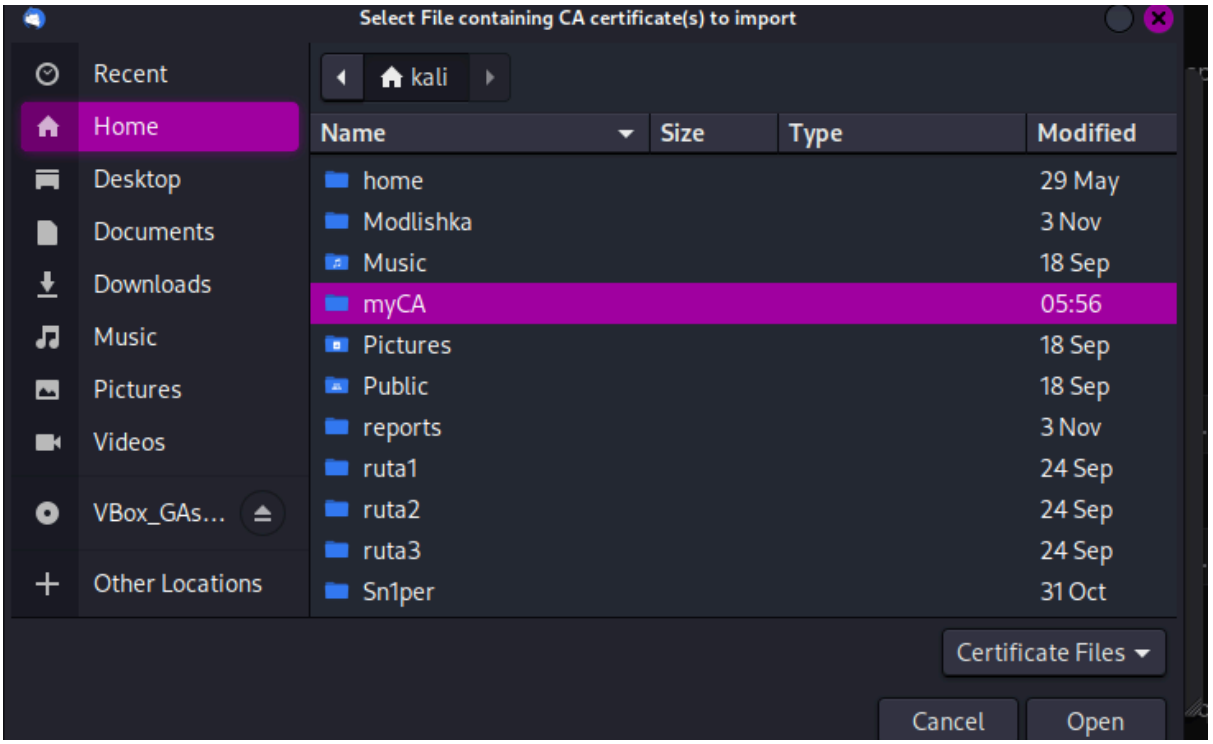
- Abre Thunderbird.
- Ve a **Editar > Configuración de la cuenta**.
- En **Seguridad**, selecciona **Gestionar certificados**.
- En la pestaña **Sus Certificados**, selecciona **Importar** y elige el archivo email.p12.
- Introduce la contraseña establecida.
- Marca la opción **Firmar digitalmente los mensajes salientes**.

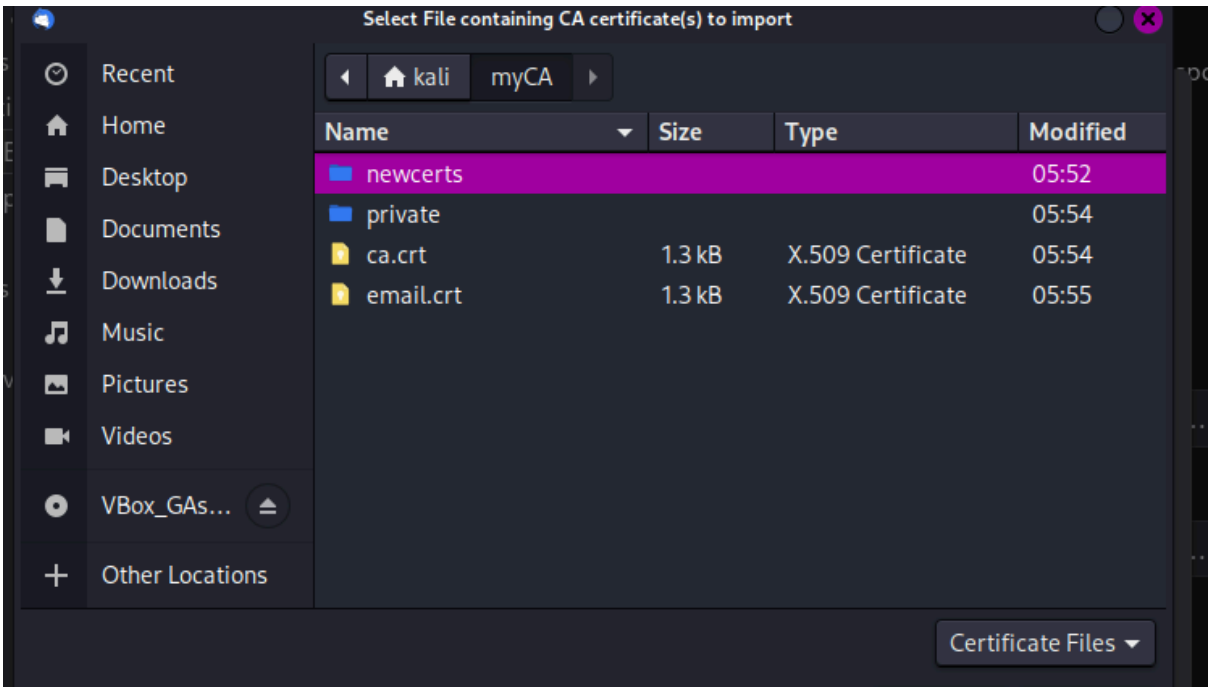

Account Setup
Thunderbird Privacy N
Set Up Your Existing Email Address
To use your current email address fill in your credentials.
Thunderbird will automatically search for a working and recommended server configuration.
Your full name
John Doe
Email address
john.doe@example.com
Password
Remember password
Cancel
Continue
Your credentials will only be stored locally on your computer.

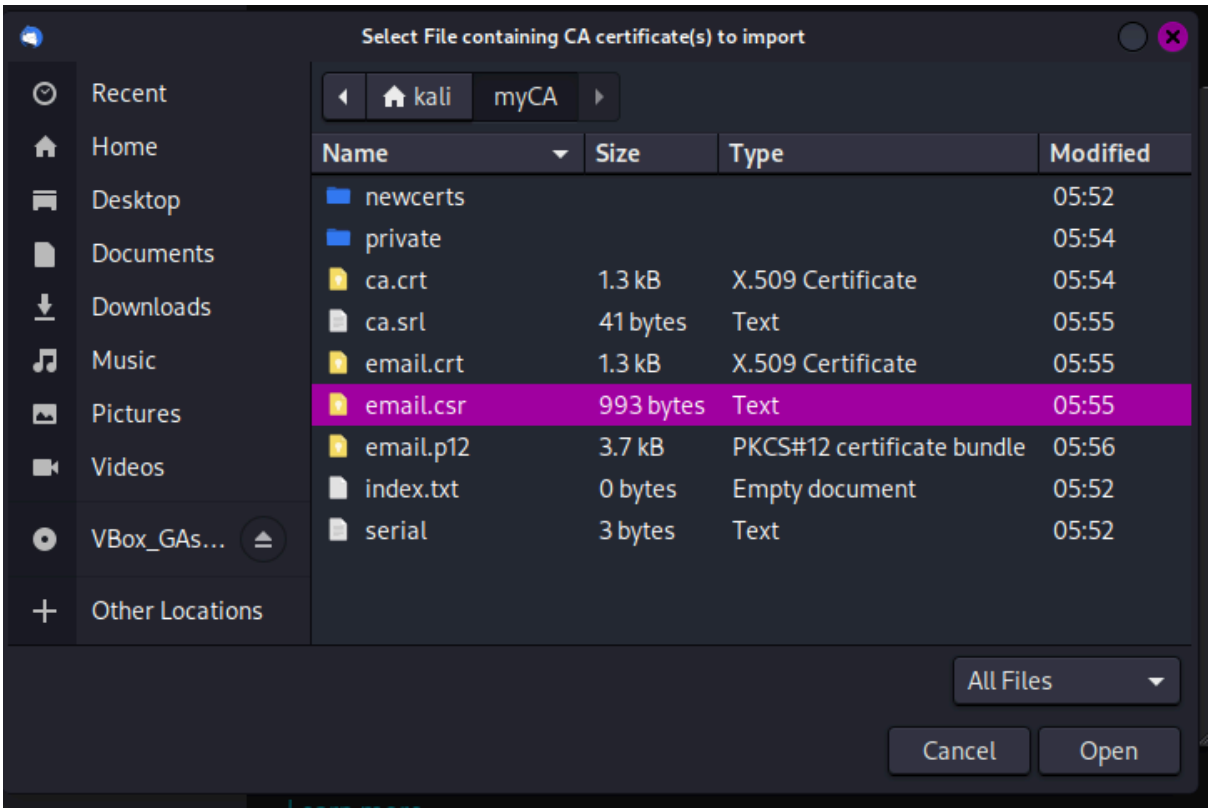

Account Setup
Thunderbird Privacy
Authentication failed. Either the entered credentials are incorrect or a separate username is required for logging in. This username is usually your Windows domain login with or without the domain (for example, janedoe or AD\\janedoe)
Configuration found in Mozilla ISP database.
Available configurations
IMAP
Keep your folders and emails synced on your server
Incoming IMAP SSL/TLS
outlook.office365.com
Outgoing SMTP STARTTLS
smtp-mail.outlook.com
Username
lorena.pki.2025@outlook.com
POP3
Keep your folders and emails on your computer
Configure manually
Cancel
Done
Your credentials will only be stored locally on your computer

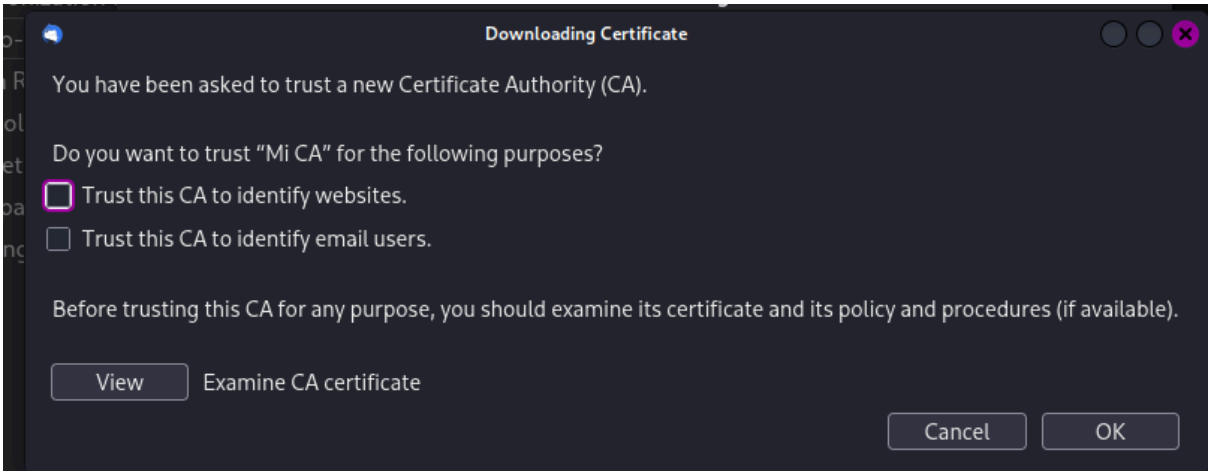

Account succes
for account
lorena.pki.2025@outlook.c
om responded: User is
authenticated but not conn
You can now use this account wit
You can improve the experience b
and configuring advanced account settings.
lorena.pki lorena.pki.2025@outlook.com
IMAP
Account settings
End-to-end encryption
Add a signature
Download dictionaries
Connect your linked services
Setup other services to get the most out of your Thunderbird
experience.
Connect to a CardDAV address book
Connect to an LDAP address book
Connect to a remote calendar
Finish

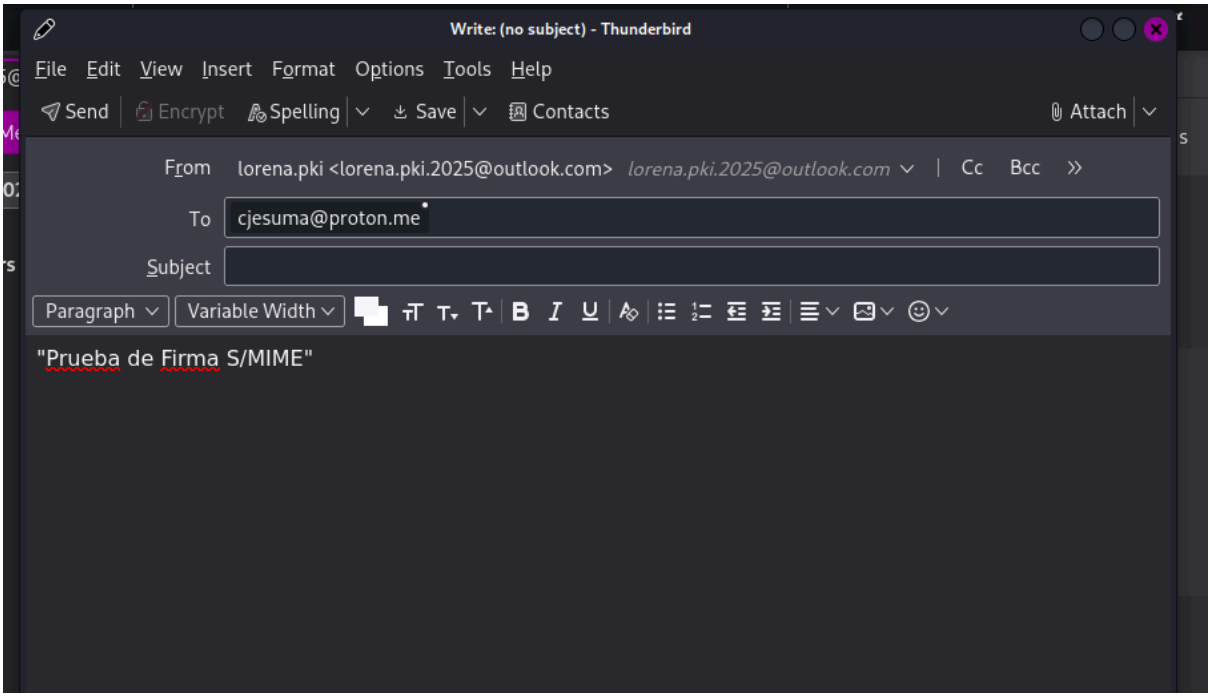

lorena.pki.2025@outlook.com
Account Settings
New Account
lorena.pki.2025@outlook.com
Server Settings
Copies & Folders
Composition & Addressing
Junk Settings
Synchronization & Storage
End-To-End Encryption
Return Receipts
Local Folders
Junk Settings
Disk Space
Outgoing Server (SMTP)
Thunderbird Settings
Add-ons and Themes
and to mass surveillance.
To send encrypted or digitally signed messages, you need to configure an encryption technology, either OpenPGP or S/MIME.
Select your personal key to enable the use of OpenPGP, or your personal certificate to enable the use of S/MIME. For a personal key or certificate you own the corresponding secret key.
Learn more
OpenPGP
Thunderbird doesn't have a personal OpenPGP key for
lorena.pki.2025@outlook.com
Add Key…
Use the OpenPGP Key Manager to view and manage public keys of your correspondents and all other keys not listed above.
OpenPGP Key Manager
S/MIME
Personal certificate for digital signing:
Select…
Clear
Personal certificate for encryption:
Select…
Clear
Manage S/MIME Certificates
S/MIME Security Devices
To obtain a new personal S/MIME certificate, generate a Certificate Signing Request (CSR) and submit it to a Certificate Authority (CA).
Learn more


Select File containing CA certificate(s) to import
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
VBox_GAs…
Other Locations
kali
Name
Size
Type
Modified
home
29 May
Modlishka
3 Nov
Music
18 Sep
myCA
05:56
Pictures
18 Sep
Public
18 Sep
reports
3 Nov
ruta1
24 Sep
ruta2
24 Sep
ruta3
24 Sep
Sn1per
31 Oct
Certificate Files
Cancel
Open


Select File containing CA certificate(s) to import
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
VBox_GAs...
Other Locations
kali
myCA
Name
Size
Type
Modified
newcerts
05:52
private
05:54
ca.crt
1.3 kB
X.509 Certificate
05:54
email.crt
1.3 kB
X.509 Certificate
05:55
Certificate Files

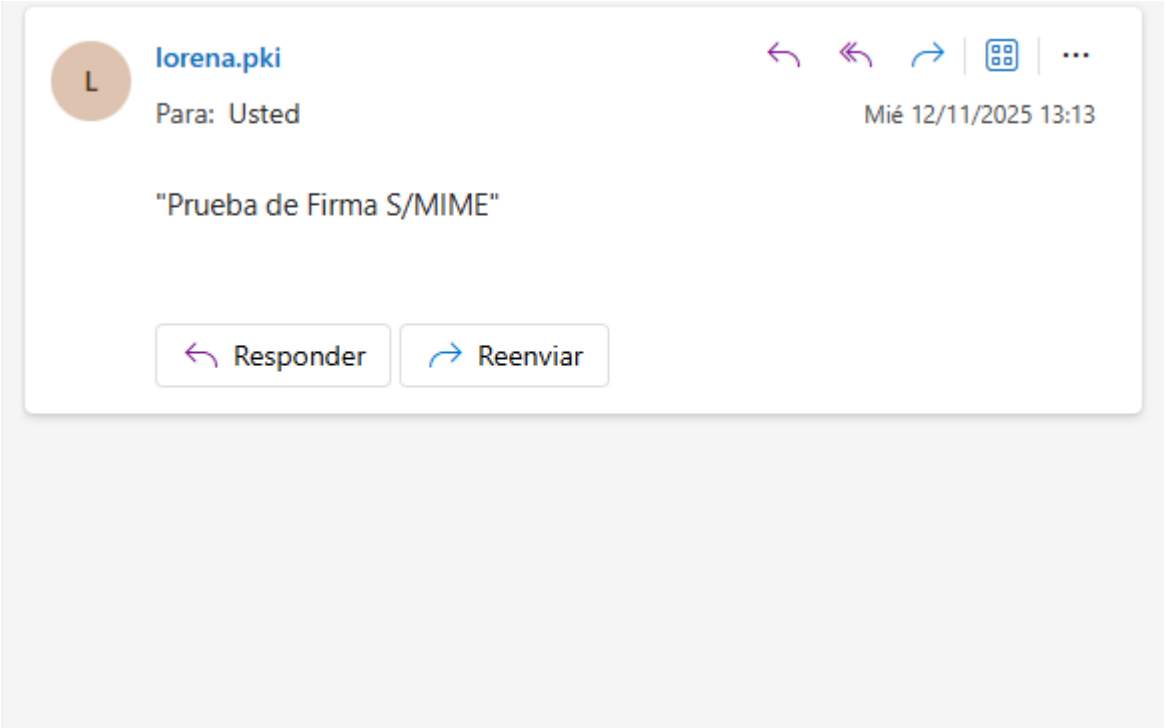

Select File containing CA certificate(s) to import
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
VBox_GAs...
Other Locations
kali
myCA
Name
Size
Type
Modified
newcerts
05:52
private
05:54
ca.crt
1.3 kB
X.509 Certificate
05:54
ca.srl
41 bytes
Text
05:55
email.crt
1.3 kB
X.509 Certificate
05:55
email.csr
993 bytes
Text
05:55
email.p12
3.7 kB
PKCS#12 certificate bundle
05:56
index.txt
0 bytes
Empty document
05:52
serial
3 bytes
Text
05:52
All Files
Cancel
Open

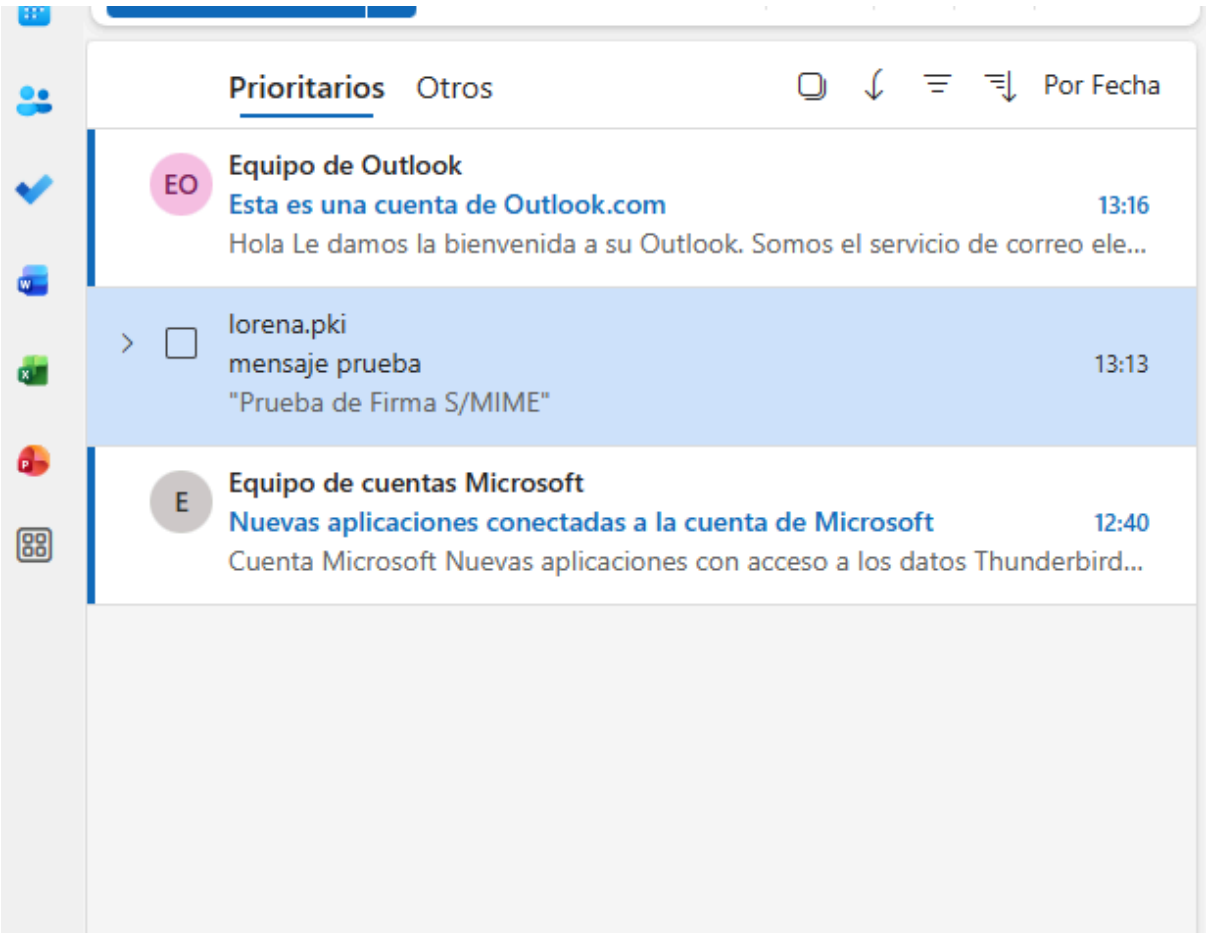




9. Probar el envío de correos firmados

- Redacta un nuevo correo en Thunderbird.
- Envía el correo a otro destinatario o a ti mismo.
- Verifica que esté firmado correctamente con S/MIME.





10. Validar la firma del correo

Cuando recibas el correo, Thunderbird mostrará un ícono indicando que el correo está

firmado.

Al hacer clic, podrás ver detalles sobre la validez del certificado.
